

Guan Chen

Summary

Platform engineer with 13+ years building distributed data infrastructure. Designed PB-scale analytical platforms, streaming systems at 10B+ events/day and 1M+ connected vehicles, and privacy computing pipelines with query planning, storage abstraction, and multi-source data access.

Experience

Senior Data Platform Engineer

Shanghai Puxin Future Internet Research Institute | Dec 2024 – Present

MIRA / ChainWeaver — Privacy Computing Platform

- Built Data Service and storage abstraction supporting 10+ data sources, billion-scale table reads, and S3 object storage access
- Designed PQL execution planning pipeline and query planner, separating query semantics from execution strategy across distributed DAGs
- Established platform interfaces and standards across planner, scheduler, execution engine, and data service components
- Built benchmark platform driving execution planning optimization with measurable evidence

Senior Data Platform Engineer / Technical Architect

Shanghai Jiayu Intelligent Robotics | Jun 2020 – Sep 2024

Connected Vehicle Telematics — National-Scale Streaming Platform (China)

- Scaled Apache Flink platform from 10,000 to 1,000,000+ connected vehicles with stable production output; 10B+ events/day on 1000+ vCore cluster
- Designed Kafka ingestion boundary decoupling vehicle T-Box producers from downstream stream processing and storage consumers
- Built Flink aggregation and enrichment pipelines; Apache Doris as real-time warehouse for business-facing analytical queries
- Established metadata and data governance as platform-wide standards across national-scale telemetry datasets

Senior Data Platform Engineer

Ping An Securities | May 2017 – Apr 2020

Enterprise Financial Analytics — Metric Platform (China)

- Reduced 1000+ vCore batch computation time from 2.5 hours to 1.5 hours through Hive query optimization and Hadoop workload tuning

- Architected metric platform serving 100,000+ searchable fields via Elasticsearch, eliminating direct Hive access for business applications
- Designed dynamic schema expansion mapping 2000+ Hive columns to Elasticsearch through a Map-based translation layer
- Established compute-serve separation: batch computation in Hive, interactive search in Elasticsearch

Data Platform Engineer — Shanghai Jiayin Data Technology

Restaurant intelligence and commercial Wi-Fi analytics; Storm, Hive, Hadoop

Research Engineer — Shanghai Ship and Shipping Research Institute

Hydrodynamic simulation workloads on Hadoop and Hive

Skills

Platform: Distributed Systems, Execution Planning, Storage Abstraction, Metadata, Data Governance

Streaming & Compute: Kafka, Flink, Spark, Hive, Hadoop

Storage & Search: HDFS, Doris, Hudi, Elasticsearch, S3

Infrastructure: Linux, Docker, Kubernetes | **Languages:** Java, Python, C

Education

Master, Naval Architecture and Ocean Engineering — Shanghai Ship and Shipping Research Institute |
Bachelor, Hydraulic Engineering — Tianjin University